

ABSTRACT

This report presents the design and implementation of a desktop-based Lost and Found Item Management System developed using Java Swing, JDBC, and MySQL. The primary objective of the system is to provide an efficient and organized platform for managing lost and found items, reducing manual effort and improving the tracking and retrieval of misplaced belongings.

The application supports essential operations such as adding, viewing, and deleting item records through a graphical user interface. Each record contains details such as item name, description, and status (lost or found). The system is structured using a layered architecture comprising Data Transfer Object (DTO), Data Access Object (DAO), Service, and User Interface (UI) layers. This separation of concerns ensures modularity, simplifies maintenance, and allows for future scalability of the application.

The DAO layer is responsible for handling database operations using JDBC, enabling secure and efficient interaction with the MySQL database. The service layer incorporates business logic and input validation to ensure the consistency and reliability of stored data. The UI layer, implemented using Java Swing, provides an intuitive interface that allows users to interact with the system and perform operations with ease.

This project demonstrates the integration of a Java-based graphical interface with a relational database system and highlights the importance of structured software design in building reliable and user-friendly item management applications.